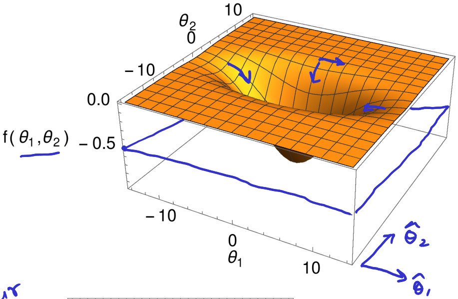
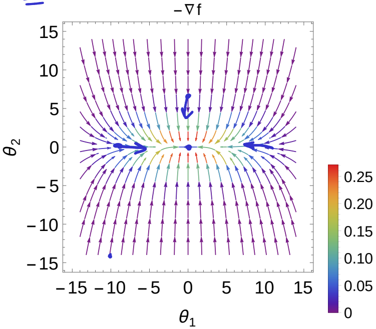
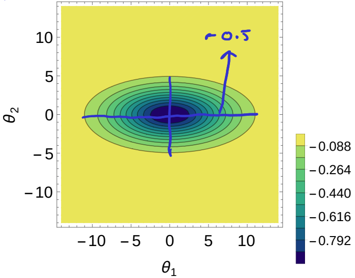
Real-valued continuous functions $f : \mathbb{R}^2 \to \mathbb{R}$





$|-\nabla f|$

Contour



$$f(\theta_1,\theta_2) = \overbrace{-\exp\left(-\frac{(\theta_1)^2+5\,(\theta_2)^2}{50}\right)}^{\mathcal{N}(\theta,\,\theta_2)}$$

grad

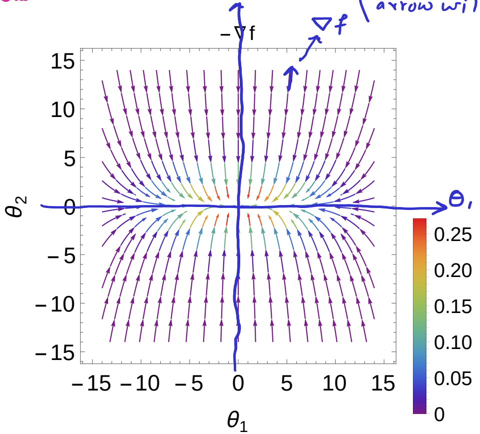
$$\nabla f = \begin{pmatrix}\frac{\partial f}{\partial \theta_1}\\ \frac{\partial f}{\partial \theta_2}\end{pmatrix} = f \times \begin{pmatrix}-\frac{2\,\theta_1}{50}\\ -\frac{10\,\theta_2}{50}\end{pmatrix}$$

θ_2 cst

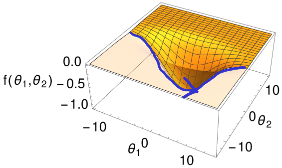
θ_1 cst

Gradient

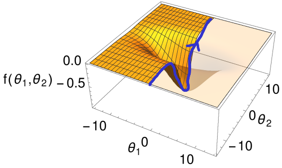
for ∇f, only dir of arrow will be flipped



$\frac{\partial f}{\partial \theta_1}$



$\frac{\partial f}{\partial \theta_2}$



$-\nabla f$ is in decreasing dir of f^n values
∇f " " Increasing " " " "

Arrows: Gradient points in the direction of steepest slope. ← fastest rate

Color bar: Its magnitude $\underbrace{|\nabla f|}_{|-\nabla f|}$ measures the steepness. ← rate

W. Rudin, *Principles of mathematical analysis* (McGraw-Hill, 1976), Chapt 9.